

Fall 2025 CPTS530 Homework 4
MATH 448-548
Numerical Analysis

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Problem 1

Problem Statement:

Solution:

Problem 2

Problem Statement:

Solution:

Problem 3

Problem Statement:

Solution:

Problem 4

Problem Statement:

Solution:

Problem 5

Problem Statement:

Solution:

Problem 6

Problem Statement:

Solution:

Appendix

This appendix contains the complete codebase for the problems solved in this homework. All implementations are written in Julia and developed from scratch without using built-in numerical solver shortcuts.

Problem 1: Code and Figures

Problem 2: Code and Figures

Problem 3: Code and Figures

Problem 4: Code and Figures

Problem 5: Code and Figures

Problem 6: Code and Figures